

7.5.4 Rotations of Conics Worksheet

1. Identify and sketch the conic given by the equation $4x^2 - \sqrt{3}xy + 3y^2 - 5 = 0$

2. Identify and sketch the conic given by the equation $2\sqrt{3}xy - 2y^2 - \sqrt{3}x - y = 0$

3. Show that the given equation is a parabola, then find its vertex, focus, and directrix in terms of the xy -axes and sketch a graph.

$$6x^2 + 4\sqrt{3}xy + 2y^2 - 9x + 9\sqrt{3}y - 63 = 0$$

4. Identify and sketch the conic given by the equation $153x^2 - 192xy + 97y^2 - 30x - 40y - 200 = 0$